Lecture 4 · Welfare, Integrability and Afriat

MWG Ch. 3.H–3.J · integrability, welfare in dollars, and finite data

Advanced Microeconomics 1

Monday 5 October

Today: run the machine backwards, then cash it out

1. **Integrability (3.H)**: which demand functions could have come from a rational consumer? Answer: exactly those with symmetric, NSD Slutsky matrices — and we can *recover* the preferences.
2. **Welfare (3.I)**: with $e(p, u)$ recoverable, "how much is this price change worth in dollars?" becomes a well-posed, answerable question — CV, EV, deadweight loss.
3. **Finite data (3.J)**: Afriat's theorem — GARP is *equivalent* to utility maximization on any finite dataset. The theory behind your Practical A code.

Integrability

The inverse problem

Integrability problem (MWG 3.H)

Given a candidate demand $x(p, w)$ — continuously differentiable, h.d.0, satisfying Walras' law — does there exist a rational, continuous, LNS preference generating it? If so, construct it.

- Strategy: don't hunt for u directly; hunt for $e(p, u)$. If we can find an expenditure function consistent with x, its upper contour sets $\{x : p\,x \geq e(p, u)\ \forall p\}$ rebuild the preference.
- By Shephard, e must solve the system of PDEs

$$\nabla_p\, e(p) = x\big(p,\ e(p)\big), \qquad e(p^0) = w^0.$$

Symmetry is the integrability condition

- A system $\nabla_p e = x(p, e)$ is locally solvable iff cross-partials commute (Frobenius). Differentiating: $\partial^2 e/\partial p_k \partial p_\ell$ computed both ways must agree — which is precisely

$$s_{\ell k}(p, w) = s_{k\ell}(p, w) : \quad \textbf{Slutsky symmetry}.$$

- NSD then guarantees the recovered e is *concave* in p, i.e. a legitimate expenditure function supporting a preference.

Proposition · summary

$x(p, w)$ (c.d., h.d.0, Walras) is generated by rational LNS preferences $\iff$ $S(p, w)$ is symmetric and negative semidefinite everywhere.

- The scorecard from L2/L4 closes: symmetry $+$ NSD is not just necessary — it is the whole story.

Why integrability earns its keep

1. **Falsifiability**: consumer theory's complete empirical content on differentiable demand is: h.d.0, Walras, S symmetric NSD. Nothing else. A theory you can write on one line is a theory you can reject.
2. **Welfare from demand**: estimation gives you $x(p, w)$; integrability is the license to convert it into $e(p, u)$ — the object welfare statements are made of. Every CV you have ever seen computed from an estimated demand curve leans on this section.
3. **Parsimony**: to specify a valid demand system, you may specify e (or v) and differentiate — automatic coherence. This is exactly how AIDS-type systems are constructed.

Welfare: CV and EV

Money-metric utility

- Ordinal utility levels mean nothing; *dollars do*. Fix reference prices $\bar{p}$ and define the **money metric** $m(\bar{p}; u) = e(\bar{p}, u)$: wealth needed at $\bar{p}$ to reach utility u.
- $e(\bar{p}, \cdot)$ is strictly increasing in u, so this is itself a valid utility representation — one calibrated in currency.
- All of applied welfare economics is: pick a reference price vector, measure everything through e.

CV and EV

Price change $p^0 \to p^1$, wealth w fixed; $u^0 = v(p^0, w)$, $u^1 = v(p^1, w)$.

Definition · compensating and equivalent variation

$$\mathrm{CV} = w - e(p^1, u^0) \qquad\qquad \mathrm{EV} = e(p^0, u^1) - w$$

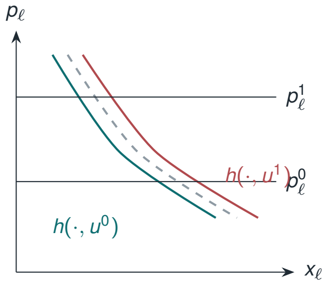
- **CV**: after the change, how much wealth can be *taken away* (given, if negative) to return the consumer to u^0? Compensation at *new* prices.
- **EV**: before the change, what wealth change is *equivalent* to it — delivers u^1 at *old* prices?
- Both are money-metric utility differences; they differ only in the reference price vector. For a price *increase* both are negative.
- Which to use? EV compares *multiple* policy scenarios on a common ruler (p^0); CV answers “what transfer implements the status quo utility” — the compensation question.

CV and EV as areas under Hicksian demands

Single price change in good ℓ: $p_\ell^0 \to p_\ell^1$. Using Shephard, $\partial e/\partial p_\ell = h_\ell$:

$$\mathrm{CV} = \int_{p_\ell^1}^{p_\ell^0} h_\ell(p, u^0)\, \mathrm{d}p_\ell, \qquad \mathrm{EV} = \int_{p_\ell^1}^{p_\ell^0} h_\ell(p, u^1)\, \mathrm{d}p_\ell.$$

- Two Hicksian curves — one per reference utility; the Walrasian curve runs between them (normal good).
- Hence for a price rise of a normal good: $|\mathrm{EV}| \le |\Delta \mathrm{CS}| \le |\mathrm{CV}|$.
- Marshallian consumer surplus is an *approximation* squeezed between the two exact answers.



- Quasilinear preferences: no wealth effects on good $\ell \Rightarrow$ the three curves coincide and $\mathrm{CV} = \mathrm{EV} = \Delta\mathrm{CS}$ exactly. Willig (1976): with small budget shares/wealth

Deadweight loss is a compensated object

- Commodity tax t on good ℓ: consumer faces $p^1 = p^0 + t\, e_\ell$; revenue $R = t \cdot x_\ell(p^1, w)$.
- **DWL (EV-based)**: $-\mathrm{EV} - R\ =$ what the consumer loses beyond what the government collects:

$$\mathrm{DWL} = \int_{p_\ell^0}^{p_\ell^0+t} \left[h_\ell(p, u^1) - h_\ell(p^0 + te_\ell, u^1)\right] \mathrm{d}p_\ell \ \geq 0 \quad \text{— the Hicksian triangle.}$$

- It is the *compensated* elasticity that scales distortions: income effects transfer resources, substitution destroys them. Second-order in t (Harberger): small taxes are cheap, doubling a tax roughly quadruples its DWL.
- This is the formal backbone of the bunching literature's welfare claims (L2): the elasticity recovered at the kink is the compensated one that belongs in this triangle.

Welfare applications at the frontier

Corners as measuring devices: bunching

Application · Saez (AEJ: Economic Policy, 2010)

- Recall the kinked budget sets from Lecture 1. At a convex kink z^* (marginal tax rises $t_0 \to t_1$), theory predicts an atom of taxpayers *exactly at* z^*: everyone whose interior optimum under t_0 lies just past the kink piles up on it.
- The excess mass, relative to the counterfactual density, maps one-to-one into the compensated elasticity of taxable income: more bunching $\iff$ more responsive agents. No instrument, no structural estimation — just the density of earnings around z^*.
- And the compensated elasticity is precisely the one that belongs in the Hicksian DWL triangle we just drew: **bunching is a deadweight-loss pipeline**, from a corner of the budget set to Harberger's formula.

Bunching in administrative data

Application · Chetty, Friedman, Olsen & Pistaferri (QJE 2011); Kleven (ARE 2016)

- Danish administrative records: sharp bunching at large kinks, strongest where frictions are low (the self-employed); attenuation elsewhere reveals *adjustment costs* — a friction parameter alongside the elasticity.
- **Notches** (discontinuous jumps in the budget set) are stronger medicine: they create a strictly dominated region, and the mass observed *inside* it measures frictions directly.
- The assumption audit is pure Ch. 2–3: what is the budget frontier? which compensation concept? what fills the counterfactual density?

Two questions you already answered

One version of the quiz asked

"What is the most you would **pay** per year to keep access to [good]?"

The other version asked

"What is the least you would **accept** to give up [good] for a year?"

- In the language of this lecture: WTP is a **CV**-flavored question, WTA an **EV**-flavored one — *the same expenditure-function difference, evaluated at different reference points*. Theory says the gap between them is a wealth effect: for goods this size, small.
- Your data, frames randomized within person across four goods: [insert: median WTP vs. median WTA by good; within-person estimate of the log gap and its CI].
- The gap you produced is [insert: a multiple of] what any plausible wealth effect can deliver.

What the gap means — and why it matters for the next slide

- The WTA/WTP disparity is one of the most replicated findings in experimental economics (Kahneman–Knetsch–Thaler 1990), largest exactly for non-market goods with no reference price — though procedures and training shrink it (Plott–Zeiler 2005). Readings: endowment effect, reference dependence.
- Within our theory the two numbers should be near-identical; outside it, *which one you elicit is a modeling decision with first-order consequences*.
- Keep that in mind: the headline valuations of free digital goods on the next slide are **WTA** numbers. Same question type you just answered — and you now know, from your own behavior, in which direction that choice pushes.

What is a free good worth?

Application · Brynjolfsson, Collis & Eggers (PNAS 2019)

- Digital goods with a *price of zero* contribute (almost) nothing to GDP — yet their variation delivers enormous utility. Measure it with the tools of 3.I: massive online discrete-choice experiments eliciting willingness to accept losing access.
- Median 2017 US valuations (per year): search engines $\approx$ \$17,500; email $\approx$ \$8,400; digital maps $\approx$ \$3,600; social media only $\approx$ \$300.
- Proposal: a welfare-adjusted growth measure ("GDP-B") that adds the consumer surplus of new and free goods to the accounts.

- Note the theory doing the work: these are *equivalent-variation* style questions — $e(p^0, u^1) - w$ with "access" as the good. The survey is the expenditure function's oracle.

New goods and the price index

Application · Hausman (1996); Boskin Commission (1996)

- A new good is a price fall from its *virtual price* (the price at which demand is zero) to its market price — valued, again, as an integral under a compensated demand curve.
- Ignoring entry of new goods and substitution biases the CPI *upward*; Boskin's estimate: about $+1.1$ percentage points per year at the time.
- Stakes: indexed benefits, tax brackets, real-wage series — decades of compounding.

- Common thread with the previous slide: whether the good's price is zero or the good is new, the missing object is the same — the area under h_ℓ where markets don't record transactions.

Finite data: Afriat

From derivatives to datasets

- Integrability assumed we *know the whole function* $x(p, w)$. Real data: T observations $\{(p^t, x^t)\}_{t=1}^T$. What is the exact empirical content of utility maximization now?
- Revealed preference relations on the data:
  - $x^t\ R^D\ x$ iff $p^t x^t \geq p^t x$ (directly revealed weakly preferred)
  - $x^t\ P^D\ x$ iff $p^t x^t > p^t x$ (directly revealed strictly preferred)
  - R = transitive closure of R^D.

Definition · GARP (Varian 1982)

The data satisfy GARP if: $x^t\, R\, x^s \implies$ not $x^s\, P^D\, x^t$.
(No revealed-preference chain may be closed by a strict arrow back.)

Afriat's Theorem

Theorem · Afriat (1967); Varian (1982)

For a finite dataset $\{(p^t, x^t)\}_{t=1}^T$ with $p^t \gg 0$, the following are equivalent:

1. the data are rationalized by some LNS utility function;
2. the data satisfy GARP;
3. there exist numbers U_t and $\lambda_t > 0$ solving the linear inequalities

$$U_s \;\leq\; U_t + \lambda_t\, p^t \cdot (x^s - x^t) \qquad \forall\, s, t;$$

4. the data are rationalized by a **continuous, increasing, concave** utility function.

- (1) $\iff$ (4) is the shocker: on finite data, concavity and monotonicity are free — they add no testable restrictions beyond GARP. "The consumer is rational" and "the consumer maximizes a nice concave utility" are observationally identical.

The construction (what your code exploits)

- Given the Afriat numbers (U_t, λ_t), define

$$u(x) \;=\; \min_t \;\{\, U_t + \lambda_t\, p^t \cdot (x - x^t) \,\}.$$

- A minimum of affine functions: automatically continuous, concave, increasing ($\lambda_t > 0$, $p^t \gg 0$). The inequalities in (3) are exactly what makes each x^t optimal in its own budget.
- Interpretation: λ_t is the marginal utility of wealth at observation t; the constructed u is a piecewise-linear “lower envelope” utility — the same envelope logic as L3, now run in reverse.
- Existence of (U_t, λ_t) is a linear feasibility problem $\Rightarrow$ checkable; GARP itself is checkable by transitive closure (Warshall) — which is literally the algorithm in Practical A.

CCEI, formally — and the power question

Definition · Afriat's critical cost efficiency index

For $e \in [0, 1]$, relax the revealed-preference relations to: $x^t \, R^D(e) \, x$ iff $e \cdot p^t x^t \geq p^t x$.
CCEI $= \sup\{e : \text{data satisfy GARP under } R(e)\}$.

- Bisection on e + a GARP check at each step: that is the whole algorithm.
- **Power (Bronars 1987)**: a test is only informative if a *random* chooser would fail it. Simulate uniform-random choices on your budgets; if their CCEIs are near 1, your budget design cannot detect irrationality. Budget lines must cross, a lot.
- Interpretation discipline for Practical A: CCEI near 1 means "consistent with *some* stable objective" — not welfare-relevant, not "correct." A wealth-destroying rule applied consistently scores 1. Keep that in your write-ups.

Takeaways and readings

Takeaways

- Full empirical content of rationality on smooth demand: h.d.0 + Walras + symmetric NSD Slutsky. Recover e by integrating Shephard.
- CV and EV are expenditure-function differences; CS is the approximation between them; DWL is a compensated triangle.
- On finite data, rationality $\equiv$ GARP $\equiv$ solvable Afriat inequalities $\equiv$ rationalizable by concave increasing u; CCEI makes it a continuous score.

Required: MWG 3.H–3.J.
Applications: Brynjolfsson–Collis–Eggers (2019); Saez (2010), Secs. I–II; Varian (1982), Secs. 1–2.
Thursday: Exercise Class 1 — sheet is out, board assignments posted today; the timed question reconstructs the duality square.
Monday: production. Read MWG 5.A–5.B ahead — a fast chapter *because* you now own duality.